Exercise Sheet

Task 1 – Tree-PLRU

Access sequence: 0xABC, 0xB43, 0xC6A, 0xDD4, 0xABC, 0xE2F

Execute the above sequence in a fully associative cache with 4 cache lines and with Tree-PLRU as replacement strategy. What does the Tree look like after the sequence has finished if at first the cache is empty and all tree nodes point to the left?

Task 2 – Tomasulo's Algorithm

Draw a simplified diagram depicting all necessary components for Tomasulo's algorithm for a CPU with 3 functional units and 3 spaces in each reservation station. In your own words, annotate each step and explain what is necessary for an instruction to enter each step and what to leave it.

Task 3 – Assembly

Write a RISC-V Assembly program that finds the largest power of 2 less than or equal to a given integer n. For example, if $n = 66$ the result is 64. Structure your program as follows: First, write a value into $t0$, this is your n. Compute your output and also write it into $t0$. The following restrictions apply:

- $n <= 2^{32-1}$
- $n > 0$
- You may only use instructions from the RV32I Base instruction set and the M-Extension

As a preparation for the exam, write your program on paper. Remember to also document your code by writing comments. Use proper syntax for your documentation, i.e. begin all your comments with a #.